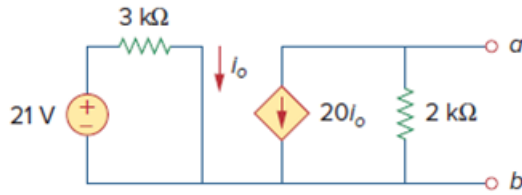


Problem

For the transistor model in Fig. 4.118, obtain the Thevenin equivalent at terminals a - b .

Figure 4.118:



Step-by-step solution

Step 1 of 6

Refer to circuit diagram in Figure 4.118 in the text book.

Consider the circuit diagram as shown in Figure 1 to calculate the Thevenin's voltage.

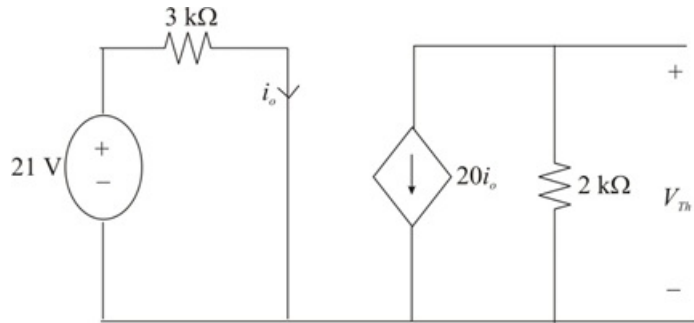


Figure 1

Step 2 of 6

Calculate the value of the current i_o .

$$\begin{aligned} i_o &= \frac{21}{3 \text{ k}} \\ &= 7 \text{ mA} \end{aligned}$$

Calculate the value of the Thevenin's voltage V_{Th} .

$$V_{Th} = (-20i_o)(2 \text{ k})$$

Substitute 7 mA for i_o .

$$\begin{aligned} V_{Th} &= (-20)(7 \text{ m})(2 \text{ k}) \\ &= -280 \text{ V} \end{aligned}$$

Therefore, the value of the Thevenin's voltage V_{Th} is $\boxed{-280 \text{ V}}$.

Step 3 of 6

Consider the circuit diagram shown in Figure 2 to calculate the short circuit current.

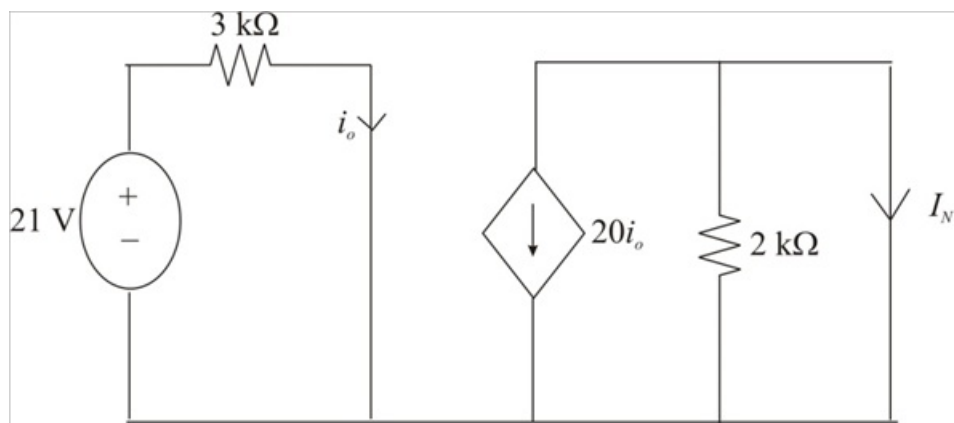


Figure 2

Step 4 of 6

No current passes through $2\text{ k}\Omega$ resistor since it is shorted.

Calculate the value of the current i_o .

$$\begin{aligned} i_o &= \frac{21}{3\text{ k}} \\ &= 7\text{ mA} \end{aligned}$$

Calculate the value of the Norton's current I_N .

$$I_N = -20i_o$$

Substitute 7 mA for i_o .

$$\begin{aligned} I_N &= (-20)(7\text{ m}) \\ &= -140\text{ mA} \end{aligned}$$

Therefore, the value of the Norton's current I_N is -140 mA .

Step 5 of 6

Calculate the value of the value of the Thevenin's resistance.

$$R_{Th} = \frac{V_{Th}}{I_N}$$

Substitute -140 mA for I_N and -280 V for V_{Th} .

$$\begin{aligned} R_{Th} &= \frac{-280}{-140\text{ m}} \\ &= 2\text{ k}\Omega \end{aligned}$$

Therefore, the value of the Thevenin's resistance is $2\text{ k}\Omega$.

Step 6 of 6

Draw the Thevenin's equivalent circuit diagram.

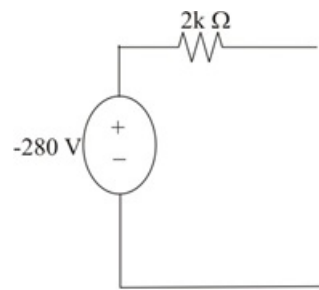


Figure 3

Therefore, the Thevenin's circuit is shown in Figure 3.